# Predicting Shipment ETA and Delay Propagation

Aiqon AI internship take-home · two-page summary. Full report: `report/report.pdf`; secondary tables and audit log: `report/appendix.md`. Every number is copied from a file generated by the code.


> **Summary.** The system predicts final ETA, the probability of a delay over 48 h, and which stage caused it, updating at five checkpoints. On **synthetic** shipments it cuts ETA error from 24.2 h (best baseline) to 16.0 h, and its explanations match ground truth 84% of the time once most stages are observed. On **real** AIS data, for the one stage that could be tested, a learned correction improves vessels' declared arrival ETA by about 1.5–2 h but barely beats a trivial baseline.


## 1. Problem, data and system

**Delay** = actual − scheduled time at a milestone; **ETA** = scheduled delivery + predicted final delay; **badly delayed** = final delay over 48 h. Predictions are made at five checkpoints (booked, departed, at port, discharged, customs cleared), each using only what is known then.

**Data.** No public dataset has scheduled and actual times for every shipment stage, so the main experiments use a causal simulator written for this project (2,000 shipments, ten India→Gulf / Singapore / Europe / US routes). Delay propagates as $d_k = m_k \cdot d_{k-1} + own_k$: slack absorbs part of the upstream delay and each stage adds its own (congestion × terminal inefficiency, missed berth windows, customs office hours, inspections, dispatch cut-offs). **Only the wind data is real**; other parameters are assumptions. Each delay hour's cause is logged, giving exact ground truth for explanations.

- **No leakage:** timelines are validated and a feature stays missing until its checkpoint; tests alter everything after a checkpoint and require the prediction there to be unchanged.
- **Models:** a *chain* model (one gradient-boosted model per stage, rolled forward: explicit propagation) blended per checkpoint with a *direct* model.
- **Uncertainty and explanations:** split-conformal 80% intervals; calibrated delay probability; hours gained or lost per stage versus a typical shipment, with the main driver and a plain-English sentence.

## 2. Results on synthetic data

Train on the earliest 60%, calibrate on 15%, test on the latest 25% (500 shipments). ETA mean absolute error, hours:

| Method | Booked | Departed | At port | Discharged | Customs cleared | Overall |
|---|---|---|---|---|---|---|
| Schedule as ETA | 52.0 | 52.0 | 52.0 | 52.0 | 52.0 | 52.0 |
| Best baseline (last delay + median remaining) | 29.0 | 29.2 | 25.8 | 21.9 | 15.1 | 24.2 |
| Direct model / Chain model | 22.7 / 27.1 | 22.8 / 26.7 | 18.8 / 17.2 | 13.9 / 12.1 | 7.0 / 4.3 | 17.0 / 17.5 |
| **Blend (this system)** | **23.1** | **23.8** | **16.9** | **11.8** | **4.2** | **16.0** |

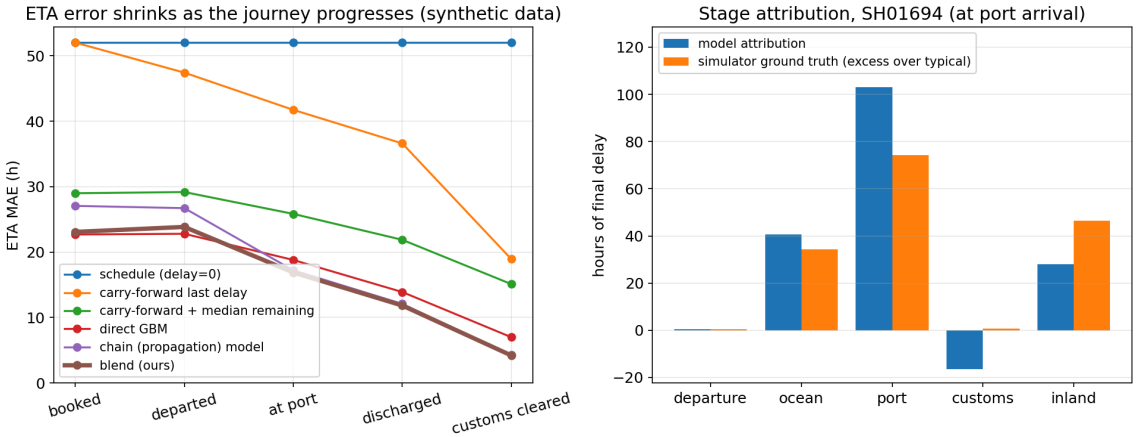



**Figure 1** (synthetic data). Left: ETA error by checkpoint. Right: delay attributed to each stage for one held-out shipment vs ground truth.

- **ETA:** 16.0 h overall (95% CI 14.7–17.3) vs 24.2 h, a paired gain of 7.1–9.5 h; 16.9 ± 1.1 h vs 25.8 ± 2.0 h across three simulated worlds.

- **Uncertainty:** 80% intervals cover 0.77–0.87 by checkpoint; delay-probability AUC 0.89 (0.82 at booking, 0.99 after customs).
- **Explanations:** the most-blamed stage matches ground truth 48% of the time at booking and 84% after customs (chance 20%).
- **Weak spots:** on two unseen routes error is 22.3 h (baseline 33.0 h) but coverage drops to 0.62–0.78; the blend beats the direct model alone by only 0.7–1.5 h (95% paired CI) and not in one of three simulated worlds; the chain is kept mainly for its stage-level explanations.

## 3. Evidence from real data

**Port-arrival ETA** was tested on 24 days of Danish Maritime Authority open AIS, where vessels broadcast a declared destination and ETA (about 30,000 vessel-hour rows, 27 ports; rows chosen with information known at that hour only, vessels that never arrive included; split by time). Error in hours on the latest 20%:

| Arrival radius | Declared ETA | Per-port median correction | Learned model | Model vs declared (95% CI) | Model vs median correction (95% CI) |
|---|---|---|---|---|---|
| 5 km | 11.46 | 7.97 | 7.41 | +2.75 to +5.41 | −0.96 to +2.14 |
| 10 km | 7.27 | 6.50 | 5.59 | +1.11 to +2.25 | +0.45 to +1.37 |
| 15 km | 5.97 | 5.32 | 4.26 | +1.28 to +2.17 | +0.76 to +1.41 |

The gain is real but modest and depends on how arrival is defined; against a trivial per-port correction the model wins by at most about an hour and at 5 km is not distinguishable. An earlier design that kept only vessels which did arrive showed a far larger gain and was **retracted**: its error scaled with the look-back window, so the selection leaked the outcome.

**Do the simulator's assumptions hold?** Tested on 1,639 real port stays (direction only): heavy-tailed port times and a late tail in arrivals are supported; the assumption that **congestion lengthens the port stage was not detected** (correlation −0.03, CI −0.08 to +0.03, vs +0.55 in the simulator), nor was a weekend effect. The proxy is crude and the ports are small, but the simulator's strongest assumption is not externally validated.

## 4. Limitations

- **Synthetic data:** the models partly recover rules I wrote; absolute errors do not predict real accuracy, and customs, delivery and end-to-end ETA are untested on real shipments.
- **Chain exposure bias:** stage models train on the true previous-stage delay but receive a predicted one. Three corrections were tested on validation data; none was better, so the bias is measured but not removed.
- **Shift:** intervals lose coverage on unseen routes; the real-data model is much weaker on unseen vessels (8.67 vs 3.08 h).

**Reproduce:** `pip install -r requirements.txt`, `python scripts/run_all.py` (about one minute), `python -m pytest tests realworld -q` (24 tests). Deterministic runs.